

Machine learning enables alignment-free distance calculation and phylogenetic placement using k-mer frequencies

Eleonora Rachtman, Siavash Mirarab

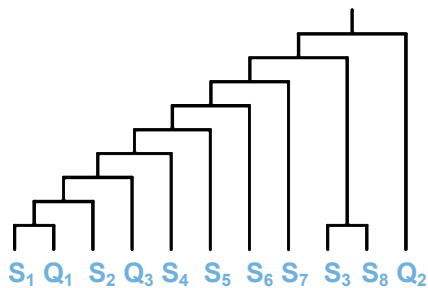
07/12/26

UC San Diego

Phylogenetic placement

S ₁	A	C	G	T	A	G	C
S ₂	A	C	G	T	A	G	T
S ₃	A	C	G	T	A	A	C
S ₄	A	C	A	T	A	G	C
S ₅	G	C	G	T	A	G	C
S ₆	A	T	G	T	A	G	C
S ₇	A	C	G	T	A	G	C
S ₈	A	C	G	T	A	G	C
Q ₁	A	C	G	T	A	G	C
Q ₂	A	C	G	T	T	G	C
Q ₃	A	C	G	T	A	G	G

de novo

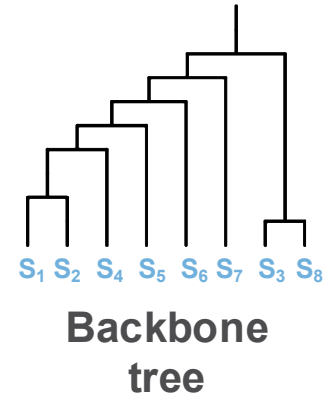
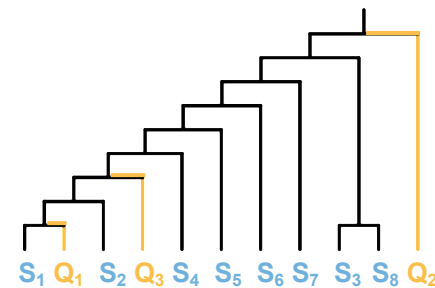


References
(backbone)

Queries

S ₁	A	C	G	T	A	G	C
S ₂	A	C	G	T	A	G	T
S ₃	A	C	G	T	A	A	C
S ₄	A	C	A	T	A	G	C
S ₅	G	C	G	T	A	G	C
S ₆	A	T	G	T	A	G	C
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Q ₂	A	C	G	T	T	G	C
Q ₃	A	C	G	T	A	G	G

Placement



Distance based phylogenetics

Sequences

H	A	T	G	C	C	A	A	G
C	A	T	G	C	C	A	A	G
G	A	T	G	C	T	A	A	G
B	A	C	G	T	T	A	C	G
S	A	C	G	T	T	G	C	G

Observed $d = 1/8$

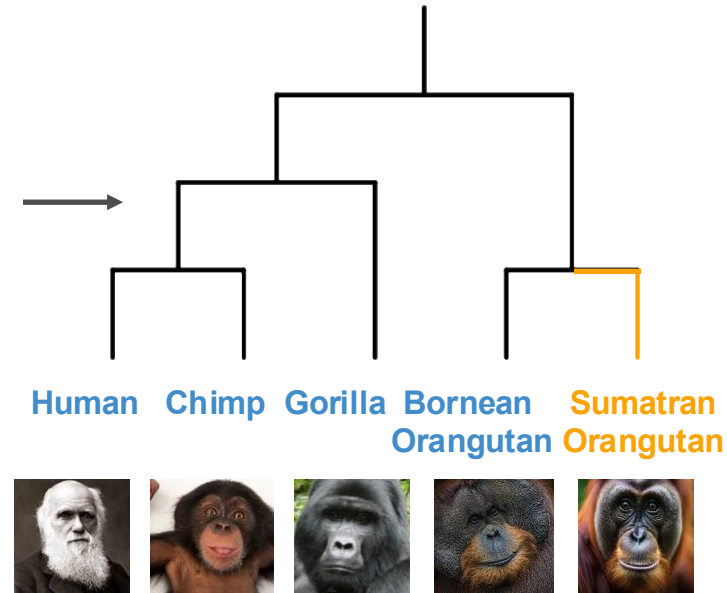


$$\text{Jukes - Cantor } d = -\frac{3}{4} \ln(1 - \frac{4}{3} d)$$

Distance matrix

	H	C	G	B	S
H	0	0.0	0.2	1.1	1.5
C		0	0.2	1.1	1.5
G			0	0.7	1.1
B				0	0.2
S					0

Phylogenetic tree



- Can we **learn a distance function** without modeling mechanistic parameters?
- Can workflow **remain alignment-free**?

k-mer based distances

➤ Long k-mers

- $k = 31$
- Absence/presence vectors
- Skmer or Mash

$$\text{Jaccard} = \frac{A_0}{A_0 + A_1 + A_2}$$

$$\text{Skmer distance} \propto \begin{cases} \text{Jaccard} \\ \text{Coverage} \\ \text{Error rate} \end{cases}$$

➤ Short k-mers

- $k < 15$
- Frequency vectors
- CAFÉ suite computes 28 alignment-free dissimilarity measures:
 - Raw k -mer counts
 - Background adjusted k -mer counts

Is there way to avoid “*human engineering*” of these distances?

Sarmashghi, S., Bohmann, K., P. Gilbert, M. T., Bafna, V., & Mirarab, S. (2019). *Biology*, 20(1), 34.
Ondov, B. D., Treangen, T. J., Melsted, P., Mallonee, A. B., Bergman, N. H., Koren, S., & Phillippy, A. M. (2016). *Genome Biology*, 17(1), 132.
Lu, Y. Y., Tang, K., Ren, J., Fuhrman, J. A., Waterman, M. S., & Sun, F. (2017). *Nucleic Acids Research*, 45(W1), W554–W559.

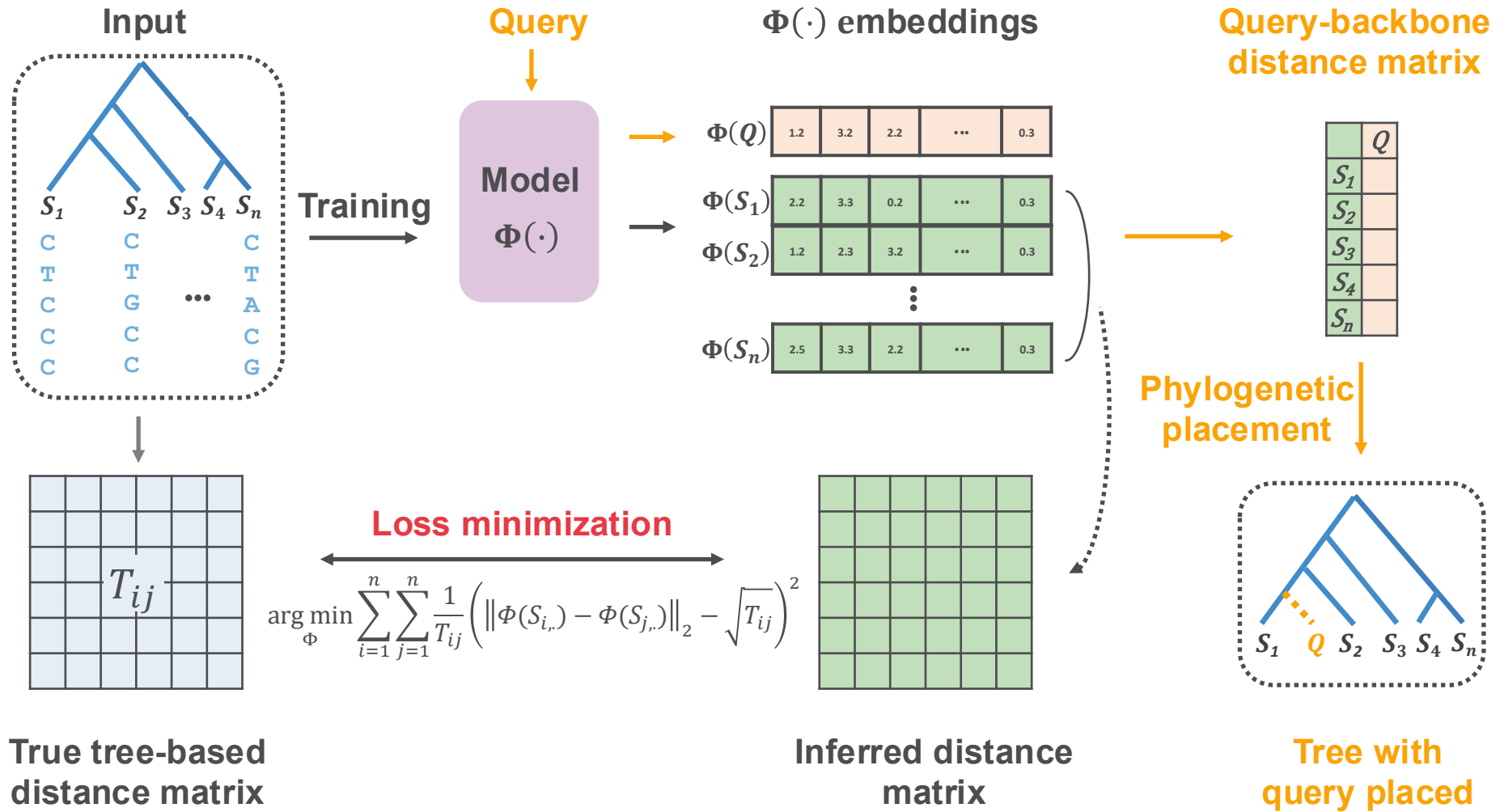


Problem definition

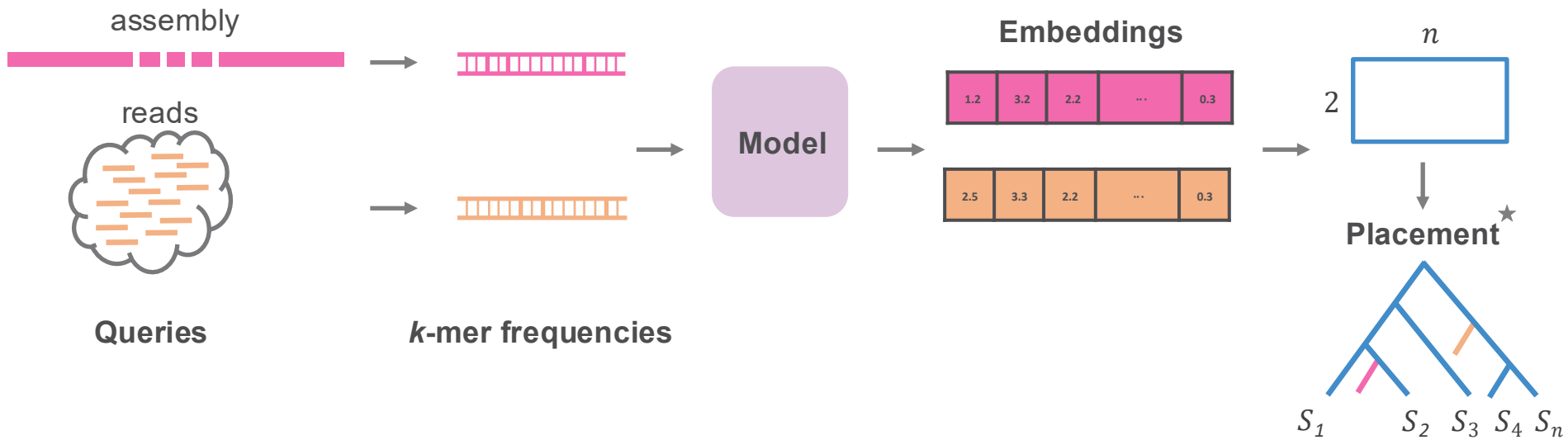
➤ Problem:

- Input: A set of sequencing reads or assemblies from each **reference and query** sample and a reference **phylogeny**.
- Output: All **pairwise distances** between queries and reference species
- Conditions:
 - Application: phylogenetic placement
 - Use machine learning to automatically learn distance function

Phylogenetic placement using machine learning



Query time



Implementation

- kf2d (k -mer frequency to distance)
 - **Placement** on bacterial/archaeal phylogeny, fungal and insect phylogeny
 - Compared multiple **model parameters** to identify best configuration
 - Evaluated **conditions** that affect placement
 - Assessed placement of symbiotic **lichens**

Molecular Ecology Resources

RESOURCE ARTICLE **OPEN ACCESS**

WILEY

Machine Learning Enables Alignment-Free Distance Calculation and Phylogenetic Placement Using k -Mer Frequencies

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<https://github.com/noraracht/kf2vec>
DOI: <http://doi.org/10.1111/1755-0998.70055>

Placement with CAFÉ

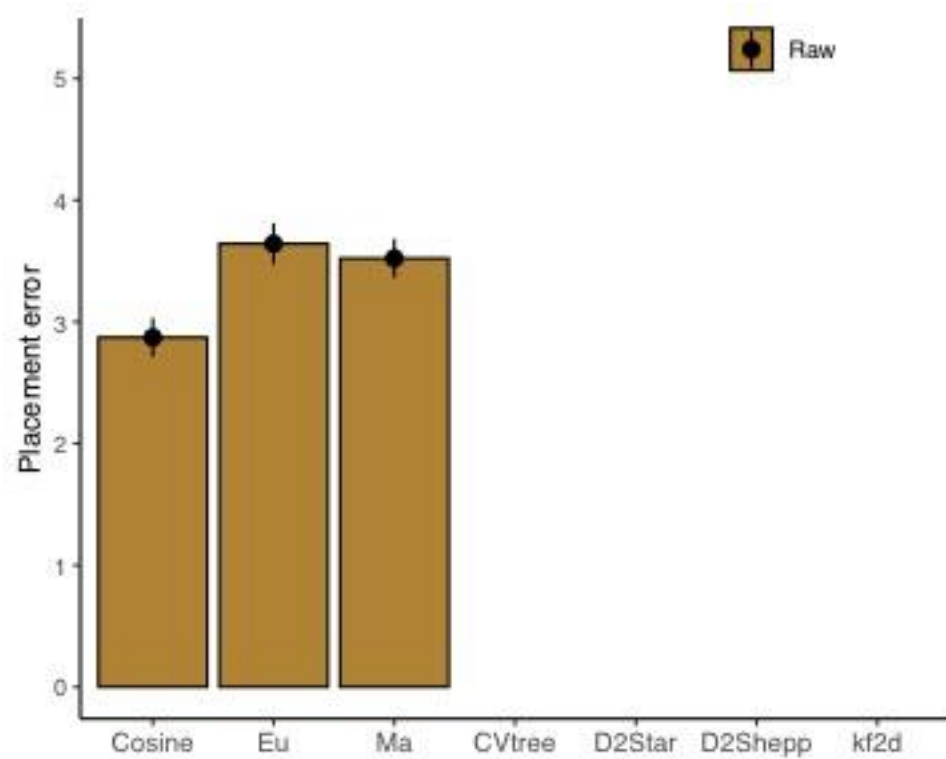
Placement error is
a count of the
number of edges
away from correct
position on
reference tree



Leave-out on
Bacterial phylogeny

Placement with CAFÉ

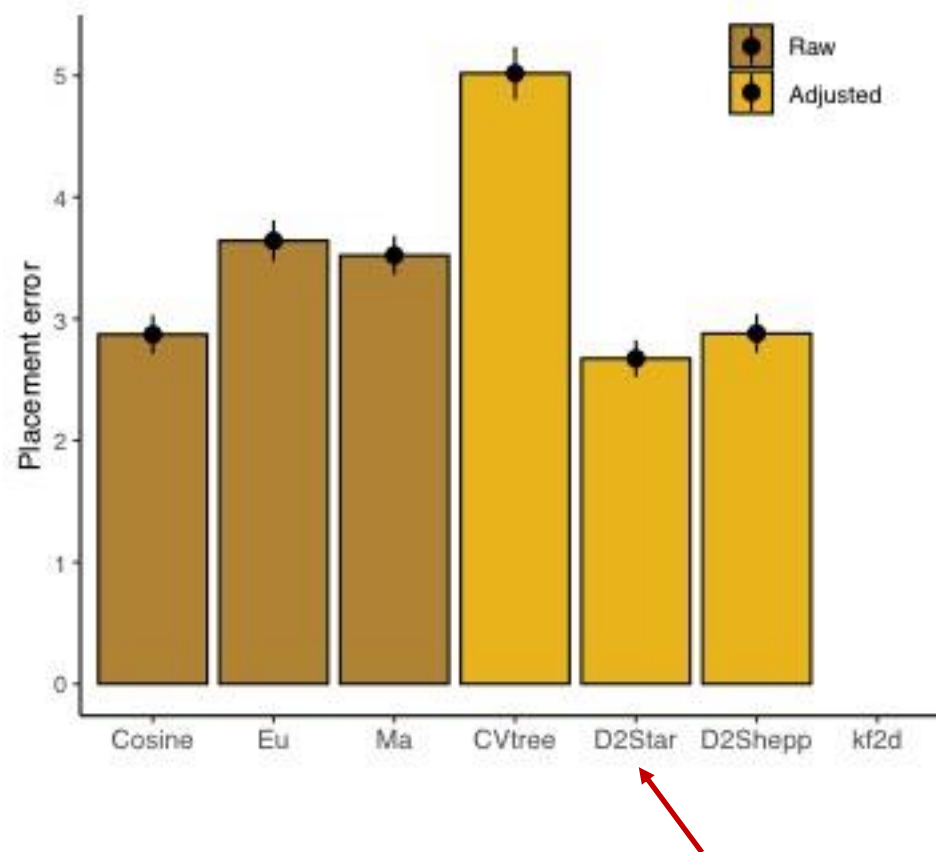
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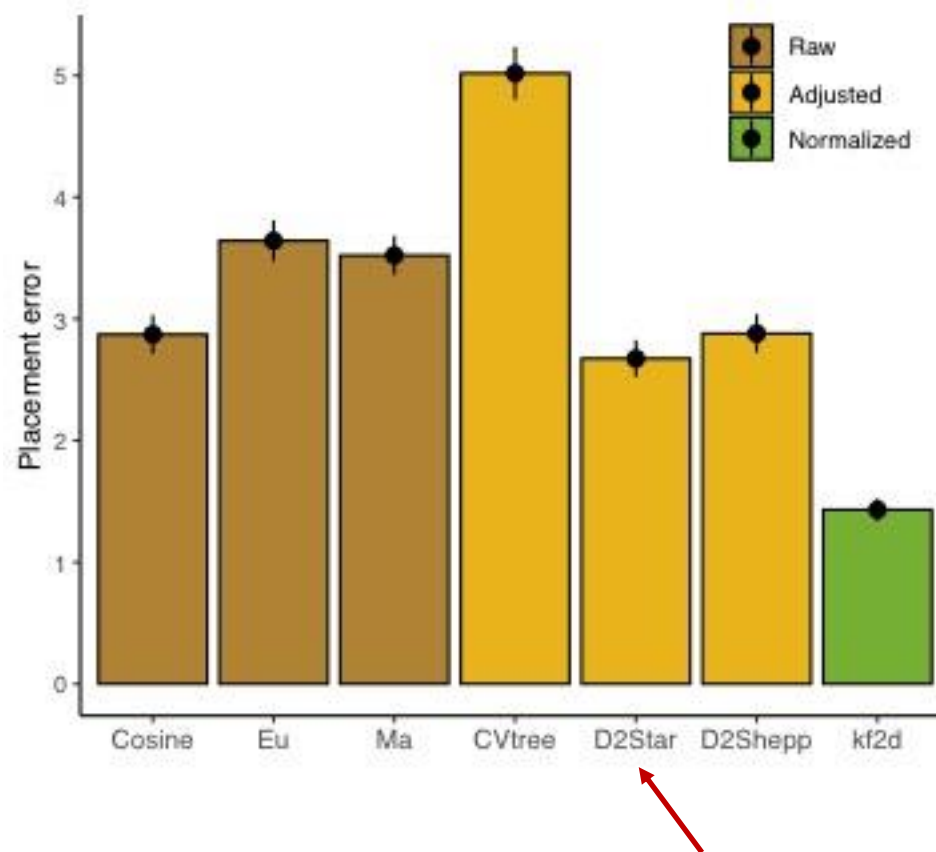
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Leave-out on
Bacterial phylogeny

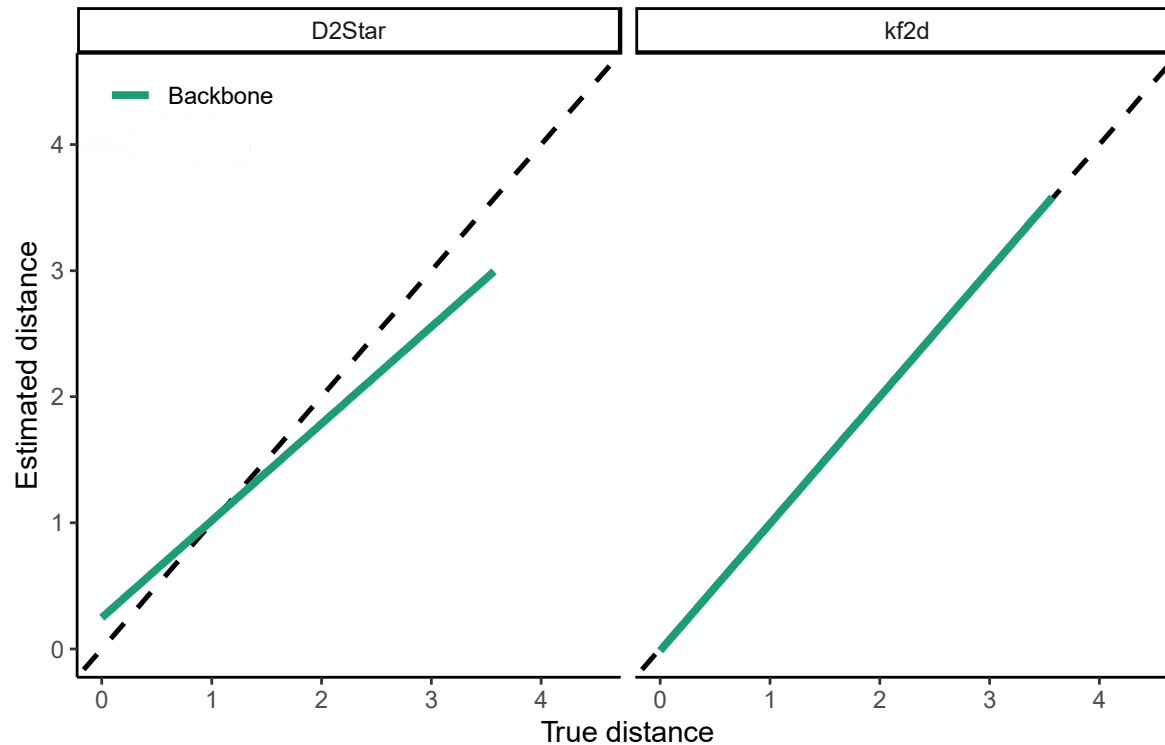
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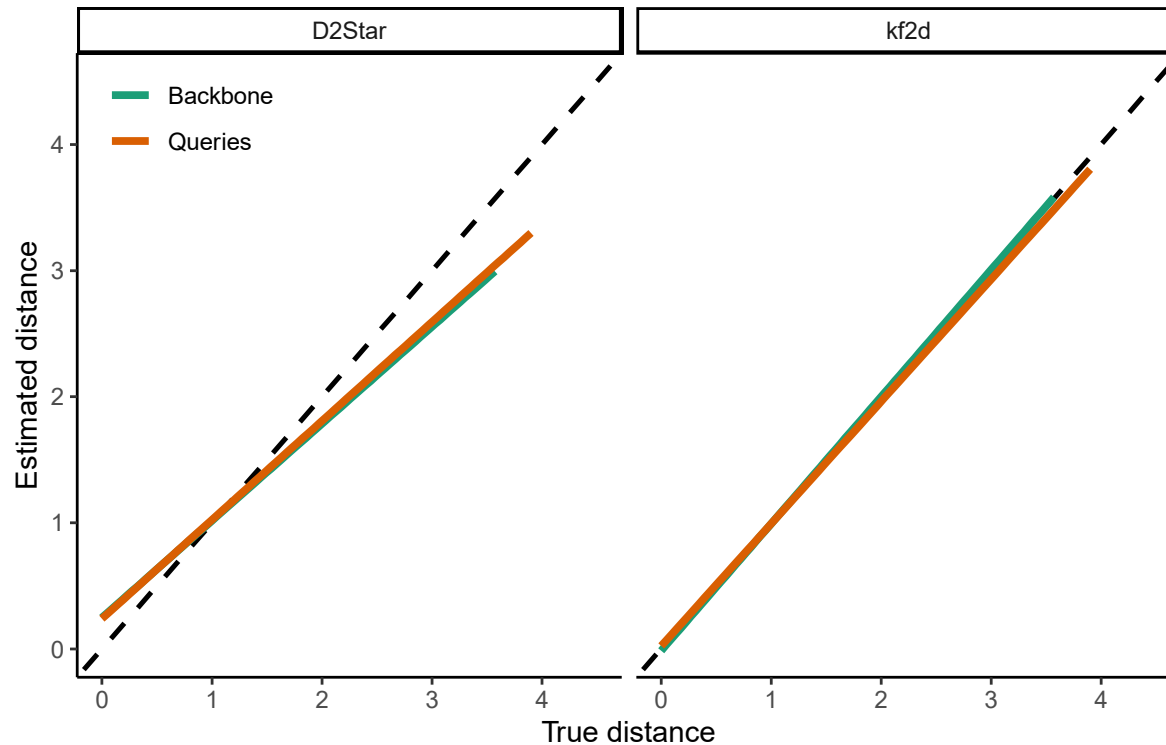
Leave-out on
Bacterial phylogeny

Distances with CAFÉ



- CAFÉ has no notion of phylogeny so let's compare estimated distances directly

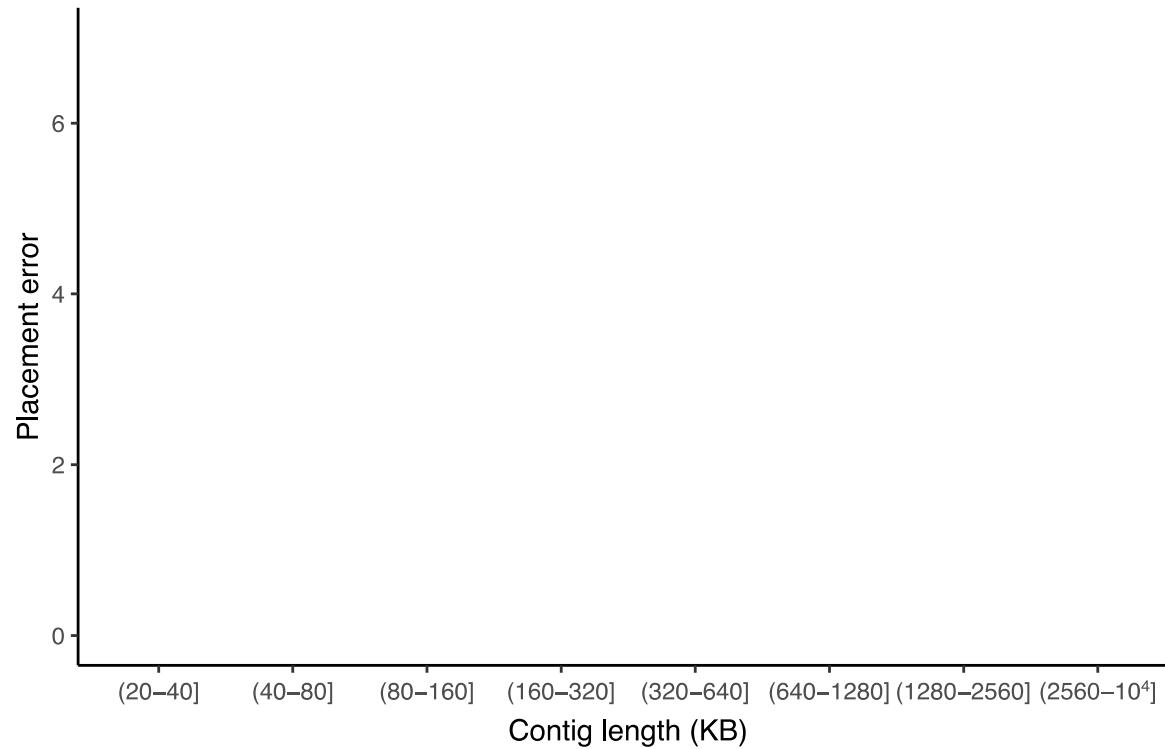
Distances with CAFÉ



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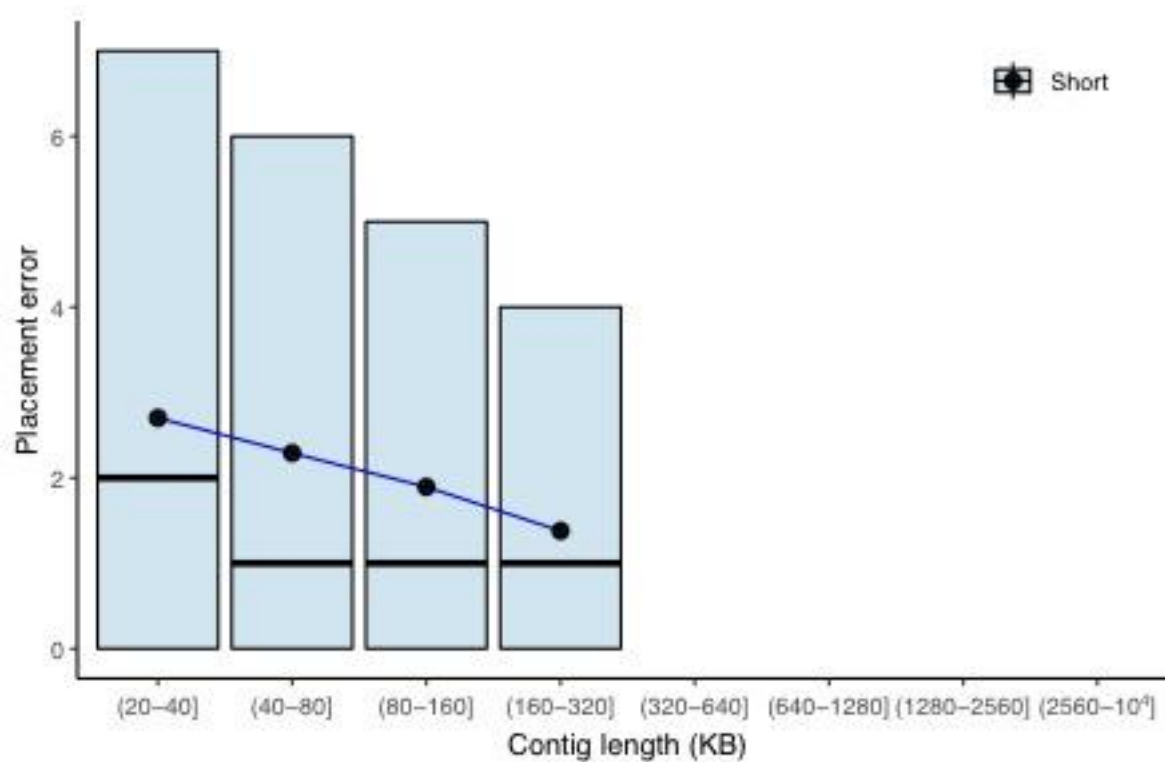
Effect of contig length

**Placing
individual
contigs**



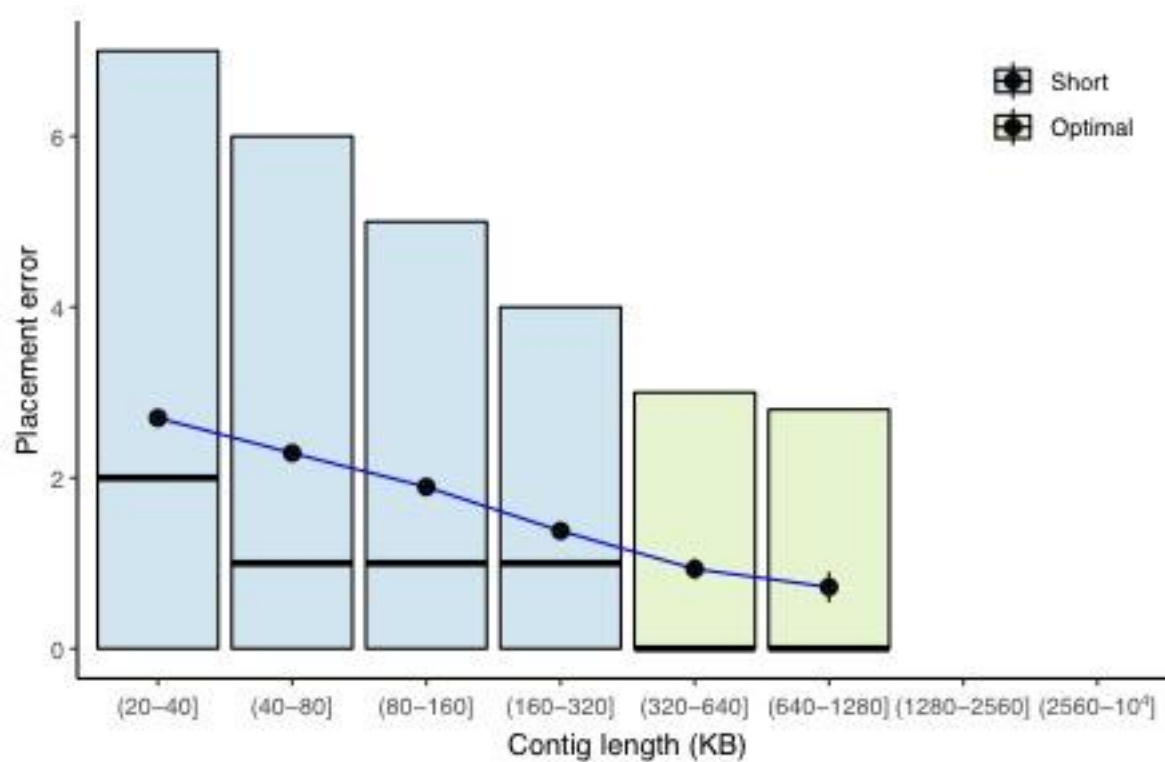
Effect of contig length

Placing
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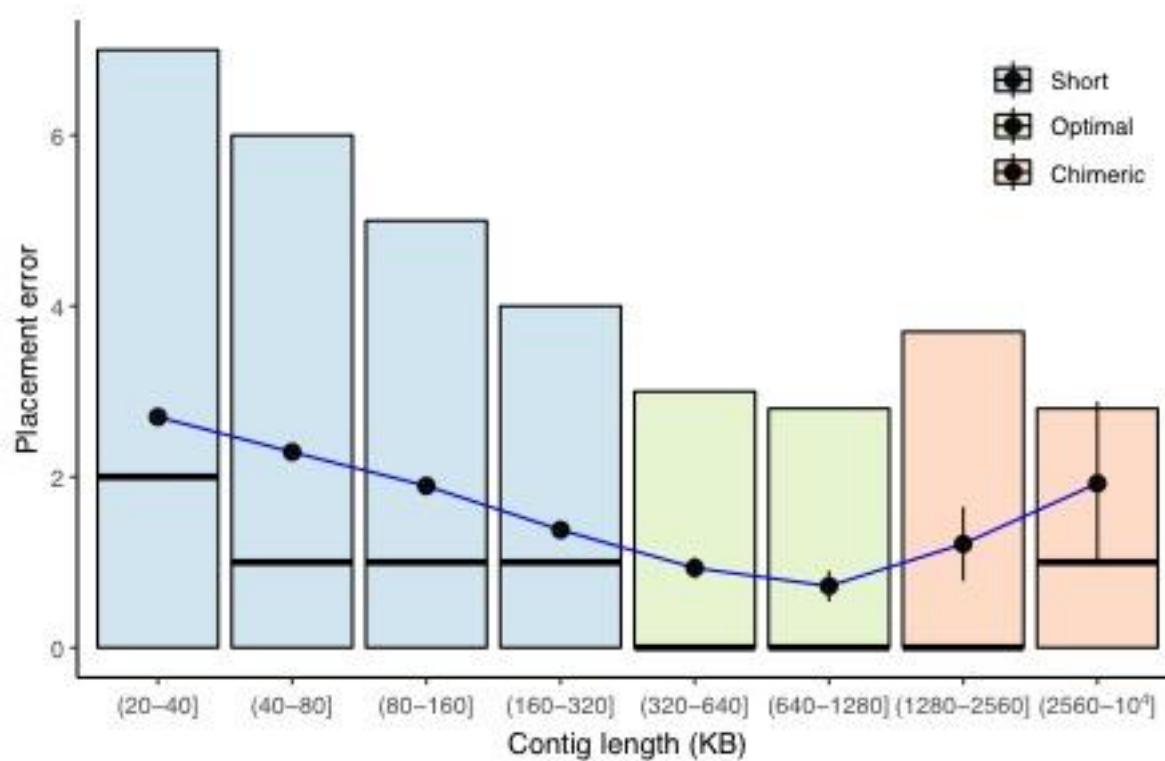
Effect of contig length

Placing
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Effect of contig length

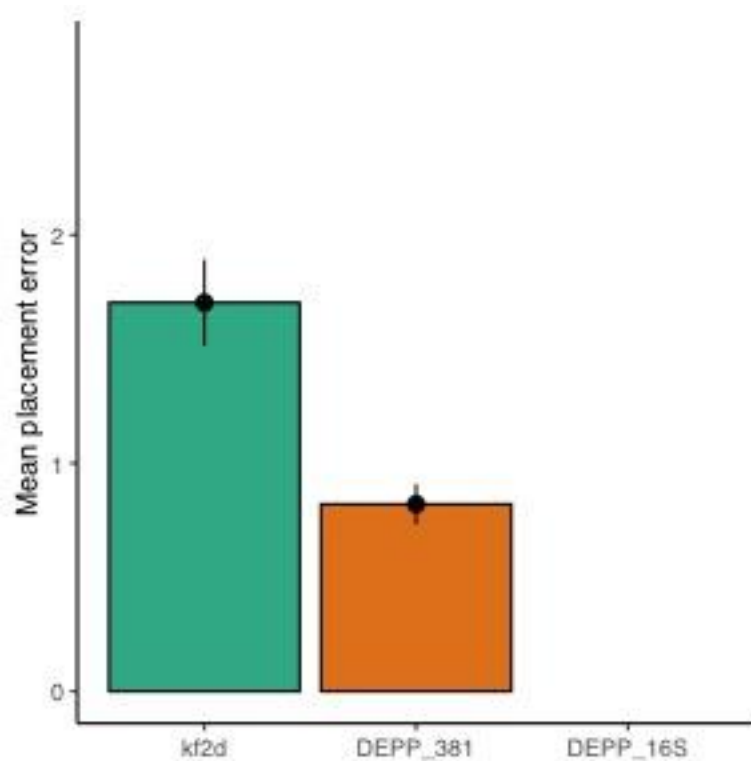
Placing
individual
contigs



Additional quality metrics

Reference representation score	Estimates how well genome is represented in the GUNC DB
Longest contig	Length of the longest contig within the genome
N50	Length for which the collection of all contigs of that length or longer covers at least half an assembly
N75	Length for which the collection of all contigs of that length or longer covers at least 75% an assembly
Mean scaffold length	Mean over lengths of all scaffolds

Comparison to alignment based

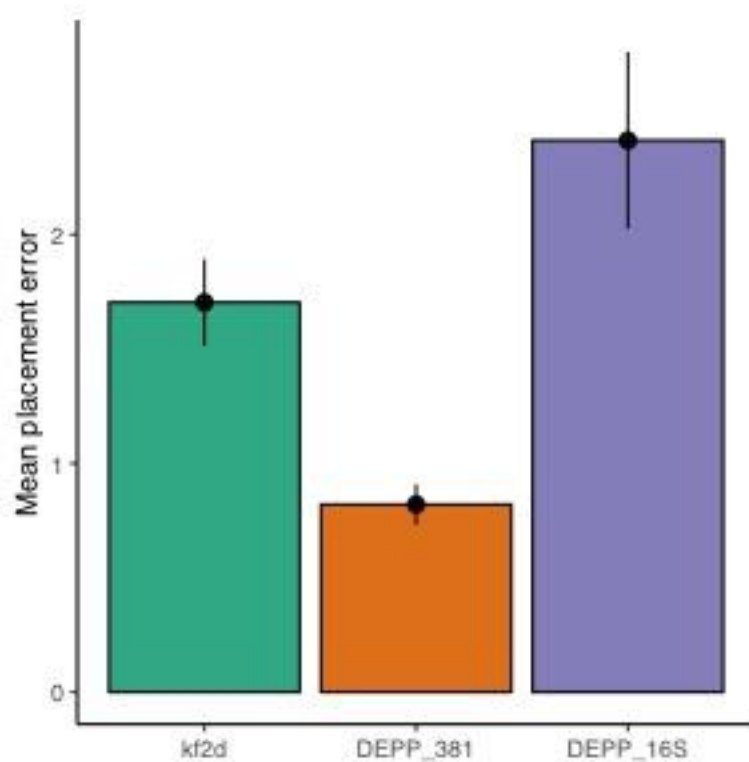


DEPP preprocessing:

- Assembly
- Identification of marker genes
- Gene annotation
- Multiple sequence alignment

How about 16S rRNA marker?

Comparison to alignment based



DEPP preprocessing:

- Assembly
- Identification of marker genes
- Gene annotation
- Multiple sequence alignment

How about 16S rRNA marker?

Performance on good quality assemblies

Dataset	Size	Query count	Placement error
Fungi	1601	73	0.05
Insects	1178	60	0.13

Li, Y., Steenwyk, J. L., Chang, Y., Wang, Y., James, T. Y., Stajich, J. E., ... Rokas, A. (2021). Current Biology, 31(8), 1653-1665.e5.
Chesters, D. (2016). Systematic Biology, syw099.

Summary

3. Machine learning enables alignment-free distance calculation and phylogenetic placement using *k*-mer frequencies
 - Interest from collaborator to try it out